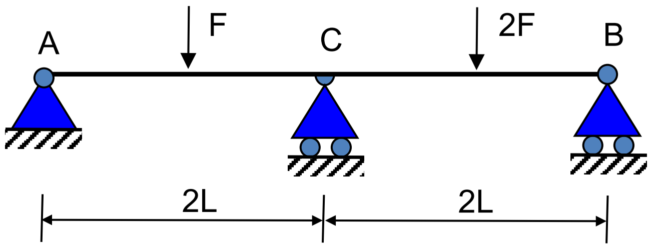
Tutorial 1 – Reactions

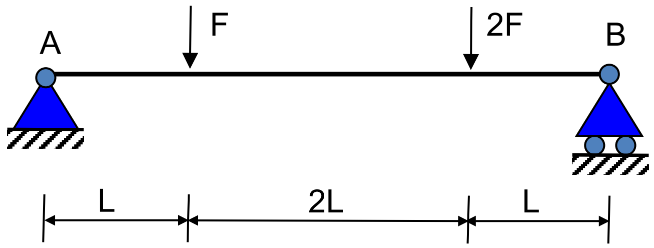
Calculation of external and internal reactions of planar structures.

- Students are expected to actively participate to tutorials, whose aim is to help students in understanding theory and preparing them for the written exam.
- Examples will be partly solved by the tutor, partly by the students themselves who can always ask for a little help when difficulties or doubts arise.
- Examples must be solved in a structured manner, drawing the graphs in scale and writing the basic step for of the solution and the final results.
- It is advisable that students work in small groups (2 or 3 persons).

Example 1:

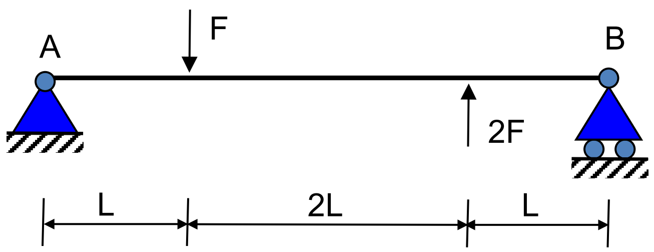


Example 2:



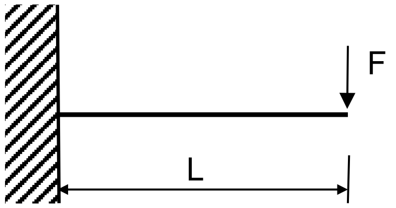
$$H_A = 0$$
$$V_A = \frac{5}{4}F$$
$$V_B = \frac{7}{4}F$$

Example 3:



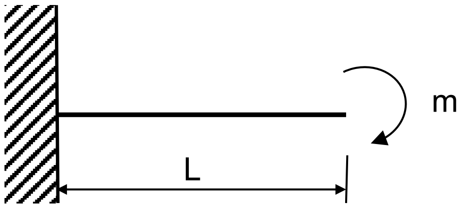
$$H_A = 0$$
$$V_A = \frac{F}{4}$$
$$V_B = -\frac{5}{4}F$$

Example 4:



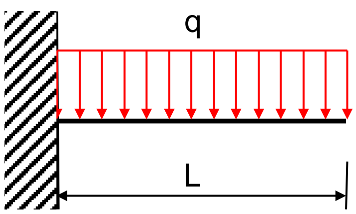
$H_A = 0$
$V_A = F$
$C_A = -FL$

Example 5:



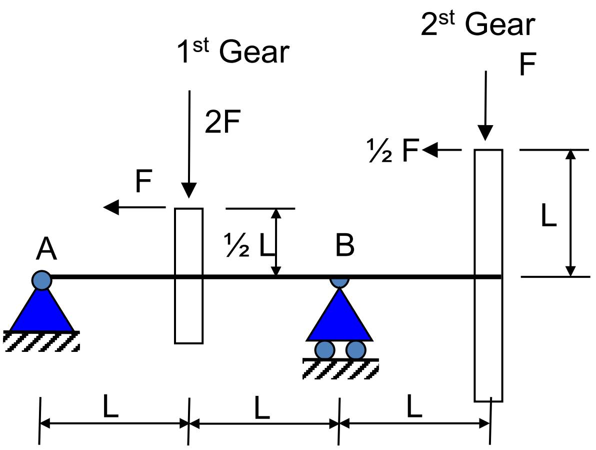
$H_A = 0$
$V_A = 0$
$C_A = -m$

Example 6:



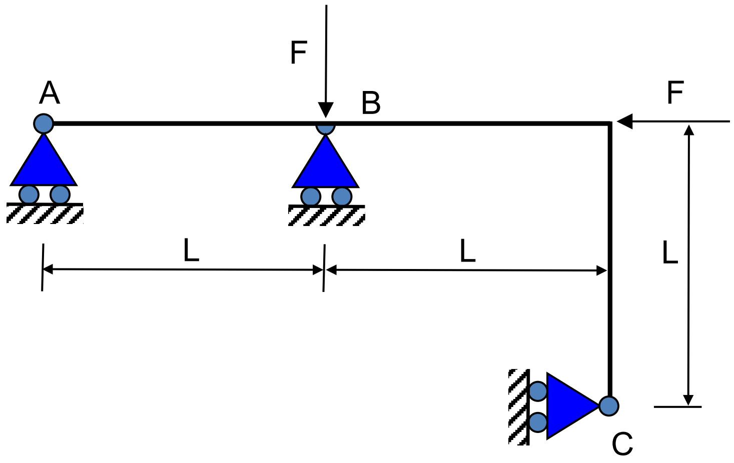
$H_A = 0$
$V_A = qL$
$C_A = -q\frac{L^2}{2}$

Example 7:



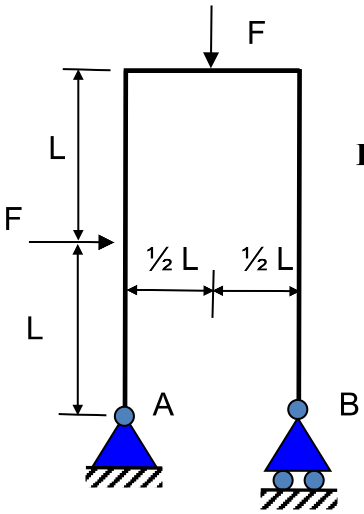
$H_A = \frac{3}{2}F$
$V_A = F$
$V_B = 2F$

Example 8:



$\boldsymbol{V_A = F}$
$\boldsymbol{V_B = 0}$
$\boldsymbol{V_C = F}$

Example 9:



Determinacy

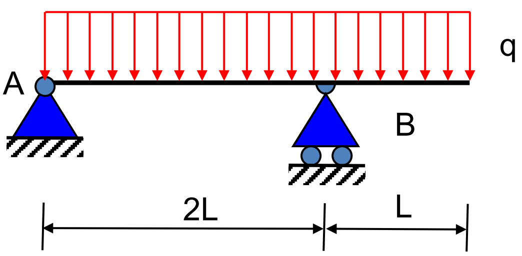
- degrees of freedom of the structure in the plane (DOF): $n = 3$
- number of DOF constrained:
  - 1 hinge = 2 degrees of freedom constrained;
  - 1 pins = 1 degrees of freedom constrained.

  $v = 1 \cdot 2 + 1 \cdot 1 = 2 + 1 = 3$

  $h = v - n = 3 - 3 = 0$ → Statically determinate problem.

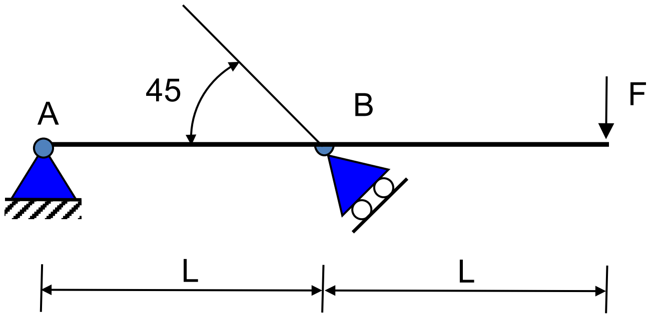
$$\boldsymbol{H_A = -F} \qquad \boldsymbol{V_A = -\frac{F}{2}} \qquad \boldsymbol{V_B = \frac{3}{2}F}$$

Example 10:



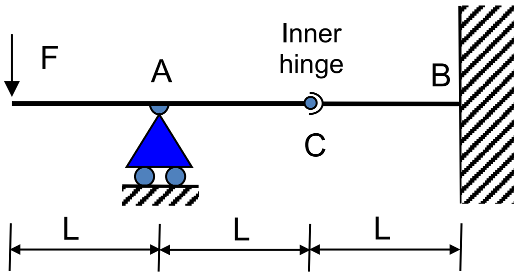
$\boldsymbol{H_A = 0}$
$\boldsymbol{V_A = \frac{3}{4}qL}$
$\boldsymbol{V_B = \frac{9}{4}qL}$

Example 11:



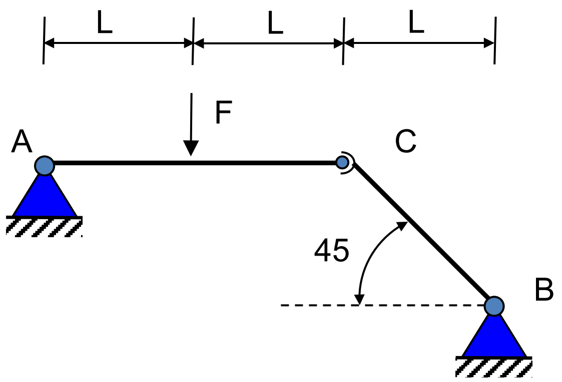
$H_A = 2F$
$V_A = -F$
$H_B = 2F$
$V_B = 2F$

Example 12:



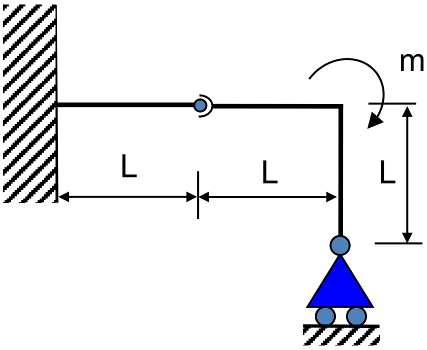
$V_A = 2F$
$H_B = 0$
$V_B = -F$
$C_B = -FL$
$H_C = 0$
$V_C = -F$

Example 13:



$H_A = \frac{F}{2}$
$V_A = \frac{F}{2}$
$H_B = \frac{F}{2}$
$V_B = \frac{F}{2}$
$H_C = \frac{F}{2}$
$V_C = \frac{F}{2}$

Example 14:



$H_A = 0$

$V_A = -\frac{m}{L}$

$C_A = m$

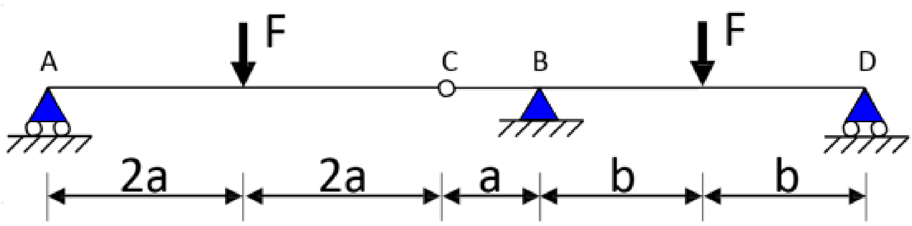
$V_B = \frac{m}{L}$

$H_C = 0$

$V_C = \frac{m}{L}$

Example 15:

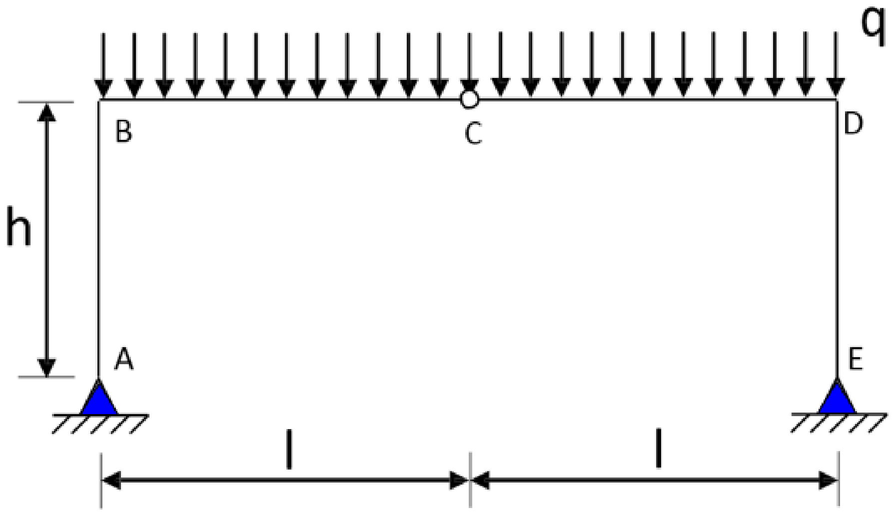
Data: (a=1.2m, b=2m, F=50N).



$V_A = 25N$ $V_B = 57.5N$ $V_C = 25\ N$ $H_C = 0$ $V_D = 17.5N$

Example 16:

Data: (l=3m, h=2m, q=20N/m),



$V_A = ql$ $\quad H_A = \frac{1}{2}\frac{q}{h}l^2$ $\quad H_C = \frac{1}{2}\frac{q}{h}l^2$ $\quad V_C = 0$

$V_E = ql$ $\quad H_E = \frac{1}{2}\frac{q}{h}l^2$